GZero 1.3 to 1.5: new analysis methods

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This note describes what changed in GZero 1.3, 1.4 and 1.5, why, and how it was checked. Version 1.3 adds three things that most break-junction papers report and GZero couldn't do yet: the junction yield with per-trace plateau statistics, single-level model fits of I-V curves, and transition voltage spectroscopy. Version 1.4 brings in the analysis from our own paper (Rashid et al., JACS 2025): the finite-temperature energy-grid fit, the noise protocol, event detection and the dwell time. It also adds two tabs for other kinds of measurements, EGaIn ensemble junctions and thermopower, and lets GZero read TDMS files of several GB. Version 1.5 adds change-point detection in breaking traces, the analysis of fixed-gap blinking (I(t)) measurements, six more file formats, and fixes for the problems found in a full review of the code.

What's new

Each addition answers a question the existing tabs couldn't.

Addition	Where	Question it answers	Module
Junction statistics	New tab after Plateau length	How many traces actually had a molecule, at what conductance, and did that drift during the run?	`gzero/junctions.py`
Single-level model fit	I-V tab	Where is the molecular level (ε_0) and how strongly is it coupled to the electrodes (Γ)?	`gzero/ivmodels.py`
Transition voltage spectroscopy	I-V tab	Where does the Fowler-Nordheim plot turn over, and what level position does that imply?	`gzero/ivmodels.py`
Events	New tab	Which traces flicker or switch mechanically?	`gzero/events.py`
EGaIn (ensemble)	New tab	What are the statistics of large-area junctions: yield, rectification and β?	`gzero/egain.py`
Thermopower	New tab	What is the Seebeck coefficient, and is transport through the HOMO or the LUMO?	`gzero/thermo.py`
Large TDMS files	Data & Traces tab	Can a recording of more than 1 GB be analysed without running out of memory?	`gzero/streaming.py`
Change points (1.5)	New tab after Junction statistics	Where does each junction rupture, what is its own plateau conductance, and which traces have more than one step?	`gzero/changepoints.py`
Blinking (1.5)	New tab after Events	In a fixed-gap I(t) measurement: at what conductance do molecules bridge the gap, for how long, and how often?	`gzero/blinking.py`

Addition	Where	Question it answers	Module
More file formats (1.5)	Data & Traces, EGaIn, Thermopower	LabVIEW .lvm, Excel, MATLAB (also v7.3), NumPy, HDF5 and gzip-compressed text	`gzero/bj_io.py`

One existing function was also fixed on the way. The averaged I-V curve used to take the overlap of all sweeps, so the half sweeps at the start and end of a recording could shrink it to almost nothing. It now only averages sweeps that cover the typical bias range.

Papers reviewed

The methods sections of the most cited break-junction papers were searched for analyses that are routinely reported but that GZero didn't have. Four papers shaped version 1.3:

Paper	What they do	What GZero does with it
Kamenetska et al., PRL 102, 126803 (2009)	Count how many breaking traces show a molecular plateau (junction formation probability) and how long the plateaus are, for a series of molecules	Junction yield per trace with a binomial error, plus yield and conductance in blocks of traces over the run
Zotti et al., Small 6, 1529 (2010)	Fit MCBJ I-V curves of tolanes with thiol, nitro and cyano anchors with a single-level model to get ε_0 and Γ	Single-level fit of every curve and of the mean forward and backward curves, with errors and an optional bias asymmetry
Beebe et al., PRL 97, 026801 (2006)	Plot ln(I/V^2) against 1/V and read off the minimum, the transition voltage V_t; it scales with the HOMO offset from photoemission	V_t for both polarities of every curve, with a Fowler-Nordheim plot of the mean curves
Baldea, PRB 85, 035442 (2012)	Shows that both polarities of V_t together give the level position ε_0 and the bias asymmetry in closed form	ε_0 and the asymmetry a from V_t^+ and V_t^-, so TVS and the fit can be compared curve by curve

The 2D correlation, clustering and flicker-noise methods from earlier papers were already in GZero. Thermopower came later, in 1.4; shot noise is still out, because it needs hardware most setups don't have.

Junction statistics

A trace counts as a molecular junction if it stays flat inside the molecular window for at least 0.1 nm, and the yield is the fraction of traces that do.

The tricky word is "flat". A trace that only tunnels still passes through a two-decade window in about 0.1-0.2 nm, so a plain length-in-window test would count almost every trace. GZero therefore looks at the local slope at each point:

$$s(z) = \frac{|\log G(z + w/2) - \log G(z - w/2)|}{w}$$

with w = 0.05 nm by default. Tunnelling falls by β / ln 10 decades per nm, which is about 9 for gold in vacuum and 4-5 in solvents, while molecular plateaus usually fall by well under 1 decade per nm. A point counts as flat below 2 decades/nm, in between the two.

For every trace this gives a flat length (the summed distance of the flat points inside the window),

a plateau conductance (the median log G of those points) and a yes or no: is the flat length at least *min plateau*? From these the tab works out the yield p with its binomial error sqrt(p(1-p)/N), the median plateau conductance and length, and the drift in decades per 1000 traces, which is a straight line through the medians of blocks of 100 traces. It plots the plateau conductance histogram, conductance and yield against trace number, the flat-length histogram with the threshold marked, and length against conductance.

In practice, Use peak from histogram fit sets the window to the fitted peak +/- 3 σ. Then have a look at the flat-length histogram: tunnelling traces pile up near zero, molecular ones form a second hump, and the threshold belongs in the dip between them.

Single-level model

The fit describes each I-V curve with one molecular level at energy ε_0 from the Fermi level, broadened by its coupling Γ to each electrode. At zero temperature the Landauer current through the resulting Lorentzian transmission has a closed form:

$$I(V) = N G_0 \Gamma \left[\arctan \frac{V/2 - \varepsilon(V)}{\Gamma} + \arctan \frac{V/2 + \varepsilon(V)}{\Gamma} \right], \quad \varepsilon(V) = \varepsilon_0 + aV$$

Energies are in eV and V in volts, so G_0 times an energy in eV is directly a current in amperes. Both electrodes are coupled equally, and N, the number of molecules in parallel, stays at 1. The parameter a lets the level move with the bias in an asymmetric junction; it is 0 unless you tick *fit asymmetry a*. At low bias the model reduces to $G/G_0 = \Gamma^2 \,/\, (\varepsilon_0^2 + \Gamma^2)$.

The current is first scaled by its largest value, so that every curve weighs the same. A least-squares fit is then started from four guesses of ε_0 (0.4, 0.8, 1.5 and 3 times the largest bias), each with a Γ that matches the measured low-bias conductance, and the one with the smallest residual wins. Errors come from the fit covariance, and R^2 is reported as well. This runs for every curve (up to 500), for the mean forward and backward curves, and for the most probable curve.

The model can't tell HOMO from LUMO transport, because both give the same I-V, so ε_0 is reported as a positive number. The thermopower tab is the way to settle the sign.

Transition voltage spectroscopy

The transition voltage V_t is where $V^2/|I|$ peaks, which is the same as the minimum of the Fowler-Nordheim plot, $\ln(|I|/V^2)$ against $1/V$. It needs no fit, and that's why so many papers report it.

For each curve and polarity GZero averages the current in 60 bias bins above 0.05 V, finds the largest $V^2/|I|$ and refines it with a parabola through that point and its two neighbours. If the maximum sits at the end of the sweep, the bias never reached V_t, and the value is left empty rather than guessed.

In the single-level model with Γ much smaller than ε_0, the two polarities give the level position and the asymmetry exactly (Baldea 2012):

$$\varepsilon_0 = \frac{2|V_t^+ V_t^-|}{\sqrt{(V_t^+)^2 + (V_t^-)^2 + \frac{10}{3}|V_t^+ V_t^-|}}, \quad a = \frac{(V_t^+ + V_t^-)\,\varepsilon_0}{4|V_t^+ V_t^-|}$$

For a symmetric junction this reduces to $\varepsilon_0 = (\text{sqrt}(3)/2)\ V_t$, about 0.87 V_t. The formula for a follows from the same stationarity condition, and it was checked numerically against the exact model (next section). Comparing the ε_0 from TVS with the one from the fit is a quick way to see whether a single level describes the curves at all.

Validation of the 1.3 methods

On data where the answer is known, every new method gets the true value back within its error. The SLM and TVS rows use model curves with 2 % noise; the yield rows use GZero's own simulator, going through the normal trace detection.

Test	True value	GZero
SLM fit, symmetric	ε_0 0.800 eV, Γ 10 meV	0.801 eV, 10.0 meV (R^2 0.9997)
SLM fit, asymmetric	ε_0 0.800 eV, a 0.100	0.795 eV, a 0.096
TVS, symmetric	ε_0 0.800 eV	V_t = +/-0.924 V, ε_0 0.800 eV
TVS, asymmetric	ε_0 0.800 eV, a 0.100	V_t^+ 1.191 V, V_t^- -0.746 V, ε_0 0.800 eV, a 0.100
TVS, asymmetric	ε_0 1.000 eV, a -0.080	ε_0 1.002 eV, a -0.080
Self-test I-V	ε_0 0.800 eV, Γ 20 meV	fit 0.800 eV, 20.0 meV; TVS 0.801 eV
Junction yield, 200 traces	70 %	68.5 +/- 3.3 %
Junction yield, 200 traces	30 %	29.0 +/- 3.2 %
Plateau conductance	log G -3.5	median -3.55

The small shift in the plateau conductance comes from the slope of the simulated plateaus (0.3 decades/nm), which pulls the median a little below where the plateau starts.

Version 1.4: methods from Rashid et al. 2025

Every analysis in the SI of our paper, Rashid et al., JACS 147, 830 (2025), is now in GZero. Only the DFT and transport calculations are not.

SI section	What the paper does	In GZero
2.3 Flicker noise	Hold 160 ms at 100 mV, drop 10 ms, keep holds whose first and last 100 points sit within +/- 1 σ of the peak, DFT squared, 100 Hz-1 kHz, n where Pearson r is smallest (0.3-2.3, step 0.01)	Flicker noise tab, preset "Rashid et al. 2025"; new *end points* setting; minimum-|r| grid value reported
2.4 I-V fit	Landauer integral with Fermi functions on an energy grid (-10 to 10 eV, 0.02 meV), $\Gamma_L = \Gamma_R$, Levenberg-Marquardt fit of the most probable I-V	I-V tab, preset "Rashid et al. 2025"; energy grid or exact digamma form; most probable curve from a Gaussian per bias bin
2.5 Mechanical modulation	2D conductance-time histograms of holds with the most probable G per time	Events tab with segments = holds
2.6 Event detection	Butterworth 200 Hz, Savitzky-Golay (order 20, 20 side points), derivative, peak thresholds 0.0007 / 0.002, width 50	Events tab
2.7 Dwell time	τ = A/w x 1000/f_s ms from the peak area	Histograms tab, with every peak fit

At finite temperature the Landauer integral is done either on the energy grid we used in the paper or with the exact closed form, which uses the digamma function psi:

$$\int T(E)\, f(E-\mu)\, dE = \pi\Gamma \left[\frac{1}{2} - \frac{1}{\pi}\,\mathrm{Im}\,\psi\!\left(\frac{1}{2} + \frac{\Gamma - i(\mu-\varepsilon)}{2\pi k_B T}\right)\right]$$

The two agree to 10^{-9}. So the fits start from the closed form and the grid only polishes the result, which brings a grid fit down from 50 s to under 1 s.

Test	True	GZero 1.4
Preset fit, model curve with the G1 values of Table S1, 300 K	ε_0 1.10 eV, Γ 11.3 meV	1.100 eV, 11.30 meV
Same, 2 % noise, through the GUI (mean, most probable, single curves)	1.10 eV, 11.3 meV	1.098-1.100 eV, 11.3 meV
Noise preset, 5 simulated sets, holds on the plateau	n = 1.5	1.53 (Morris preset: 1.59)
Dwell time, 70 % molecular, 0.4 nm at 2 nm/s	140 ms	135-146 ms

The SI doesn't give the temperature, so the preset assumes 300 K. Change it if you measured at another temperature.

Events tab

The tab finds flickering and mechanical switching in G(t) the way SI 2.6 describes, and sorts every trace into mechanical, flickering or quiet. G(t), in G_0, is smoothed with a 200 Hz, order 2 Butterworth low-pass, run forwards and backwards so the events don't shift in time, and then with a Savitzky-Golay filter of order 20 over 41 points. scipy's own coefficients collapse to about 10^{-17} at that order, so GZero builds the filter from a QR decomposition in the Legendre basis instead, which matches scipy to 10^{-14} at low order. The smoothed trace is differentiated with a central difference, and peaks of dG/dt above 0.0007 are flickers, above 0.002 mechanical events. Events while the junction is outside the molecular log G range are dropped as false positives.

The plots show an example trace with its events, its derivative with both thresholds, the share of each kind of trace, flickers per trace, and a before/after transition map. With holds you also get the conductance-time histogram with the most probable G.

Three details aren't pinned down in the SI, so GZero does what LabVIEW does by default. They're worth checking against the original VI:

- The 20 ms is LabVIEW's dt input, so dG/dt = (G[i+1] - G[i-1]) / (2 x 0.02 s). Since the dG/dt peak is about 2.2 times the step, the thresholds then correspond to steps of about 3 x 10^{-4} G_0 (flicker) and 9 x 10^{-4} G_0 (mechanical).
- "Width 50" is the window of the Peak Detector's quadratic fit, not a minimum peak width. Read as a minimum width at half height it would reject almost every step, because after the 200 Hz filter the derivative peaks are only 20-30 samples wide at 10 kHz. Both readings can be selected.
- Only rises are counted, as in the Peak Detector, so a flicker up and back counts once. *Count* = *both* counts every step.

On synthetic steps, a 4 x 10^{-4} G_0 flicker at 10^{-3} G_0 is found as a flicker, a jump from 10^{-6} to 1.6 x 10^{-3} G_0 as mechanical, and a flat trace with 3 % noise gives nothing.

EGaIn ensemble tab

Large-area junctions are analysed per junction, not per sweep: 20 junctions with 20 sweeps each are 20 independent measurements, not 400. The tab follows Reus et al. 2012, as XMe and GaussFit do.

A file is either one junction, with the bias and current columns chosen in the settings, or many junctions stored as column pairs V, I, V, I, . . . , which may differ in length. The current becomes J = I / A, with A from the contact diameter (as in XMe) or typed in. A junction whose |J| reaches the

short threshold (100 A/cm2) is a short, and the yield is working junctions over all junctions. The sweeps go onto a common bias grid, and at each bias log10|J| is histogrammed and fitted with a Gaussian, which gives the centre and $\sigma_{\log}$. The confidence interval uses the number of junctions n_j as degrees of freedom, as GaussFit does. The tab also works out log R = log|J(+V)| - log|J(-V)| and V_{trans} per cycle, and fits the single-level model to the mean J(V) with N free. Across data sets with lengths d, it fits β, weighting each point by the standard error of its mean (Simeone et al. 2013 report 0.92 per carbon):

$$\mathrm{CI} = t_{1-\alpha/2,\,n_j-1}\,\frac{\sigma_{\log}}{\sqrt{n_j-1}}, \qquad \log_{10}|J| = \log_{10} J_0 - \frac{\beta\, d}{\ln 10}$$

Check (simulated junctions)	True	Found
β, 5 sets of 30 junctions	0.9	0.904 +/- 0.016
Rectification log R at 0.5 V	0.30	0.30 +/- 0.08
Mean log|J| per bias	as simulated	inside its 95 % CI
β error coverage, 20 repeats	68 % within 1 σ	65 %

The first version weighted β by $\sigma_{\log}$ and claimed errors three times too small. Weighting by the standard error fixed that, which is what the coverage row shows.

Thermopower tab

The tab takes the thermoelectric voltage and the temperature (or temperature difference) of every sample and follows Reddy et al. 2007 and Widawsky et al. 2012. The temperature column can hold dT, or the hot-side temperature (then dT = T - T_ref), or be missing, in which case each data set gets its own dT. The samples, or the file means (one per junction), are grouped by dT, and a Gaussian is fitted to the thermovoltage histogram at every dT. A straight line through the most probable voltages gives S from its slope:

$$S = -\frac{dV}{d(\Delta T)} + S_{\mathrm{electrode}}$$

S > 0 means hole transport through the HOMO and S < 0 electron transport through the LUMO. S_i = -V_i/dT_i + S_electrode of every sample is histogrammed too, and plotted against log G when there is a conductance column. The error of each peak comes from a bootstrap over the files, because the samples within one trace aren't independent.

Check (simulated junctions)	True	Found
S, 160 traces at dT = 0-30 K	12 uV/K	11.9 +/- 0.4 uV/K
S, negative	-8 uV/K	-8.1 +/- 0.4 uV/K
Real files: V in mV, hot-side T, G window	-6 uV/K	-5.9 +/- 0.5 uV/K
Error coverage, 30 repeats	68 % within 1 σ	80 % (slightly conservative)

Large TDMS files

Converting a 1 GB TDMS file in one go needs several GB of memory, since every step makes another array as long as the file. Files above 300 MB are therefore read in pieces of 2 minutes. Loading keeps only a min/max overview for the plot. Cutting the traces reads each piece with a 60 s margin on

both sides and keeps only the traces and holds that start inside the piece, so a trace crossing a border is found once and in full. On a simulated recording this gives exactly the same traces as reading the whole file, and the self-test checks that. A 1.29 GB file (90 minutes at 10 kHz) now loads in 2.8 s and needs 0.7 GB of memory at most, against 15.6 s and 4.4 GB before.

Version 1.5: change points, blinking, file formats

For 1.5 the recent literature on break-junction analysis was searched for methods that answer questions the existing tabs can't. Two stood out. Change-point detection gives one conductance per junction, measured over its own plateau, and finds traces with more than one step. Blinking (the I(t) method) is a whole measurement mode in its own right, and GZero had no tab for it.

Change points

The rupture of each trace follows Hamill et al. 2024. Inside the window between log G = -2 and -5.5, log G against distance is fitted by one straight line and by two lines meeting at a break k. The Chow statistic (Chow 1960) compares the two:

$$F(k) = \frac{\left(S_1 - S_2(k)\right)/p}{S_2(k)/(n-2p)}, \qquad p = 2$$

The k with the largest F is the rupture, and the median log G before it is the plateau conductance of that junction. A large F alone isn't enough, because a plain tunnelling decay also bends slightly and its noise is small. So a trace only counts as a junction if the line before the break is also flat: |slope| below 2 decades/nm, the same criterion as the Junction statistics tab. On simulated traces the slope before the break is -0.3 decades/nm for every molecular trace and -4.3 for every blank one, so the two groups don't overlap.

The steps use PELT (Killick et al. 2012) for a piecewise-constant fit of log G with a penalty of 2 ln(n) per change point and the noise σ estimated from the trace (at least 0.05 decades). A variable number of change points per trace is the idea of PELT-R (Vaidya, Dyer and Monti 2026). A tunnelling ramp is cut into many short levels by such a fit, so only levels that are themselves flat and at least 0.05 nm long count as plateau levels.

Check	True	Found
Rupture, 200 traces, 70 % molecular	70 %	70.0 % junctions
Rupture, 200 traces, 30 % molecular	30 %	31.5 %
Plateau conductance per junction	log G -3.5 (spread 0.25)	median -3.55, spread 0.24
PELT against an exhaustive search, 5 random series	same change points	identical
PELT, trace with plateaus at -2.9 and -3.9 (0.3 nm each)	two levels	-2.90 and -3.90, 0.30 and 0.32 nm

Blinking

With the gap held fixed, a molecule bridging it shows up as a jump of the current to the molecular level, and its leaving as a jump back (Haiss et al. 2003, 2006). The baseline is a low percentile (20 %) of log G in 0.5 s blocks, so it follows drift without being pulled up by the blinks. A blink starts 0.6 decades above the baseline and ends below 0.3 decades (hysteresis). The blink conductance is a Gaussian fit to the blink levels. The on and off times are fitted with the maximum-likelihood lifetime of an exponential above the shortest time measured:

$$\tau = \frac{N}{N-1}\,(\bar{t} - t_{\min})\,, \qquad \delta\tau = \tau/\sqrt{N}$$

Check (simulated 20 s recordings)	True	Found
Blinks found	143, 172, 186	137, 171, 179
Blink conductance	log G -3.5	-3.51 to -3.53 (+/- 0.01)
tau_on	20 ms	19.7-23.4 ms (+/- 1.5-2.0)
tau_off	100 ms	91-121 ms (+/- 7-10)

File formats

Every non-TDMS format is turned into the same list of named columns, so the column settings work unchanged. New in 1.5: LabVIEW `.lvm` (header names, `Delta_X` as the sampling interval), Excel `.xlsx` (every sheet), MATLAB `.mat` (struct fields, v7.3 through h5py), NumPy `.npy` / `.npz`, HDF5 `.h5` / `.hdf5` (data sets by path, sampling rate from an attribute), and gzip-compressed text. The self-test writes a model trace in each of these formats, reads it back and checks that the conductance agrees to 10^{-5} decades.

Fixes

A complete review of the code found these problems, all fixed in 1.5:

Problem	Effect before 1.5
The delimiter of a text file was guessed from the first line, even if that was a header with a comma	Such files could not be read
Missing current samples (NaN) and zero bias became G = 100 G_0	Fake metal-contact spikes; now they go to the floor
The diode calibration left NaN inputs unset	Random values in the converted current
The noise power of each window was integrated after interpolation onto the frequency grid of the first window	With whole-hold windows of different length, NP depended on which hold came first; now each window is integrated on its own frequencies
A peak fit was kept when the histogram settings changed but the number of traces didn't	An old fit and dwell time could be shown and saved with a new histogram
Junction statistics, Correlation, Flicker noise, Events, I-V and Piezo modulation kept their results after Clear or new traces	Old results could be saved under the name of new data
Plateau export with no valid window	Error during Save results
The largest point fell out of the binned noise trend	Last bin one point short
The 2D histogram in `density` mode wasn't divided by the bin area	Wrong scale
A recording with fewer than 3 piezo samples	Error during trace detection
The bell fit was repeated at every n of the scan (up to 200 fits)	Very slow with few windows; now at most 40 fits, interpolated

Open-source code reviewed

Two public codes cover the same ground, and both were read in full for their EGaIn and I-V methods.

Code	Group	License	Modules	How GZero used it
XMe_DataAnalysis	Wenjing Hong, Xiamen University	Apache 2.0	Basic histograms, correlation, spectral clustering, PSD noise, I-V, single trace, thermopower, electrochemistry, EGaIn	EGaIn: J from contact diameter, log|J| per bias with Gaussian fits. The methods were re-implemented; XMe asks to be cited when compared with
GaussFit	Ryan Chiechi	GPL-3.0	EGaIn / CP-AFM parsing, Gaussian log|J|, rectification, V_{trans}, dJ/dV, lag filter, SLM	Junction-based CI, rectification and V_{trans} per cycle. Methods only: GPL code can't go into Apache-licensed GZero

Of the other Hong-group analyses in XMe, GZero already had correlation, spectral clustering and flicker noise, and thermopower is in now. Electrochemical gating needs a measurement type GZero doesn't read yet.

Sources

Code: arman-rashid/GZero, release v1.5. The full method description is in the user manual (`docs/User_Manual.pdf`), sections 3.1, 4.7, 5.2, 6.4, 6.5, 9.5, 9.8, 9.9 and 10.1-10.6, references 26-43.

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- XMe_DataAnalysis and GaussFit on GitHub